
BRUSHWORN



Too Hot to Run

Recognizing extreme heat exposure and the early signs of heat exhaustion is critical to keeping a working dog alive. A field guide for training, testing, and early-season hunts.

A BRUSHWORN FIELD GUIDE

WRITTEN BY P.J. ROSE

AND CHARLES ELLITHORPE, DVM

Why nobody can give you a number

Most of what can go wrong with a dog in the field happens without warning. Heat is different. It builds on a schedule you can read, which means the decision that matters gets made before the dog ever leaves the truck.

You would think, then, that somebody would have published the number for a temperature, a humidity reading, or a line you don't cross.

There isn't one. We looked. The veterinary literature doesn't agree, and the variables that matter, coat, conditioning, cover, airflow, acclimation and the individual dog, don't reduce to a single threshold. Anyone who hands you a hard cutoff is guessing.

So this guide does something different. It teaches you to read the conditions and the dog, because that is the skill that could save your dog's life. The rest of this document is that skill, broken into parts.

One piece of information before we get started

There is a practical rule of thumb in circulation: add the air temperature in Fahrenheit to the relative humidity level, and if the sum exceeds 150, heat injury becomes considerably more likely. Eighty-five degrees at seventy percent humidity gives you 155. Eighty-five at forty gives you 125.

Treat it as a flag, not a limit. It is a crude instrument, its relationship to the human heat index isn't clear, and it takes no account of any of the factors in the chapters that follow. But it's a five-second calculation you can do in the truck.

One caveat before we move on. We can find that rule repeated in a lot of places and cannot find where it started. Charles calls that dogma, meaning something repeated until it gets treated as fact. Take it as a simple guideline of unknown parentage.

How a dog actually sheds heat

This is the part worth understanding, because every piece of cooling advice later in this guide depends on it.

A dog sheds most of its heat by convection, which is air or water moving across the body, and by radiation, which is heat lost to the surroundings. Both start to fail as the air temperature closes on the dog's own body temperature, and panting, which cools by evaporation, has to carry more of the load.

Evaporation gets much less effective as the air gets more humid. Seventy-eight degrees at low humidity and seventy-eight degrees in hot soup are two different days for a dog, and the thermometer on the truck doesn't know the difference.

Three consequences worth remembering

- Air moving across a dog does real cooling work, more than shade alone. This is why a fan in a crate matters, and why a shaded box with no air movement can still be dangerous.
- As the air temperature climbs toward the dog's body temperature, normally 101 to 102.5 degrees, the two most efficient mechanisms quietly stop helping and the dog is left with panting.
- Humidity can significantly disable panting. That is the whole reason a humid seventy-degree day can be worse than a dry ninety.

The tools, and why each one works

Every tool at your disposal works through one of the mechanisms above, and each one has a point where it stops helping. Knowing which is which is what turns a list of tricks into a plan.

- A fan works through convection. It is the tool that keeps working when shade alone doesn't.

- Water works first by conduction and then by evaporation. That's why it does its best work with air moving across it, and why its value drops as humidity climbs.
- Shade removes the sun's radiant load and does nothing for air movement. A shaded crate in still air can still be dangerous.
- Air conditioning in the cab does all of it at once, which is why it may be one of the best tools you have.

READ THE DAY

The conditions that stack against you

No single one of these is a stop sign. They accumulate, and the more of them present at once, the smaller your margin for error. Consider them together.

Air temperature and humidity

Covered above. The two numbers mean more in combination, and humidity is the one people underestimate.

Ground temperature

Ground temperature is not air temperature, and our dogs are a lot closer to the ground than we remember. Bare dirt, gravel two-track, sand, waterside rocks and cut stubble can run far hotter than the air above them. Your dog is working in that layer, not the one you are.

The back of your hand held to the ground for a few seconds is a cruder instrument than a thermometer. Accurate or not, it gets the point across, and it puts you in the dog's zone for a moment. Do it every so often.

Thick, wet cover

Dense, saturated cover that holds water and blocks airflow is its own climate. The dog works harder to move through it, and the still, humid air inside it is where evaporative cooling is least effective.

There is no rule of thumb for how much wet cover shortens a working session. What we can tell you is that the combination is what does the damage: more humidity coming off the wet vegetation, less air

moving through it, and a wet coat on the dog holding heat in. Put those three together and it is a worse environment than an open field at the same temperature.

Time of year, and the acclimation trap

Heatstroke cases rise in spring and early summer, for reasons that compound one another. Temperatures climb faster in those seasons, the dog has had no chance to acclimate, and his fitness is often at its lowest of the year.

The practical version: the first warm week of the year is more dangerous than a hotter week in August, and early-season training is where handlers get caught.

As for building a deliberate acclimation plan, there isn't a one-size-fits-all solution, and we would rather tell you that than invent one. If it matters to your program, the people to ask are canine sports medicine specialists, and it is worth a conversation with one about your dogs and your season.

Airflow, or the lack of it

A still day is a harder day on your dog, and a crate without moving air is harder still. Without air movement, convection stops doing its share, and the dog is left with the one mechanism humidity has already compromised.

READ THE DOG

Your dog's margin

Some dogs are working with a smaller margin for error before the tailgate ever drops. These characteristics raise the risk. Treat them as a starting point, not a verdict.

- Coat and color. A dark coat absorbs more of the sun's load. Density matters as well. Know the difference between a dense coat and an open one.
- Conditioning. An unfit dog produces the same heat with less capacity to shed it, and less cardiovascular ability to recover. This cannot be overstated.
- Weight. Know what a healthy weight and body condition is for your dog. A veterinarian who isn't familiar with working dogs may not judge your dog overweight when they should. There is something to the old saying that your hunting dog should look, to your neighbor, like you're starving him. Hot, humid stretches of the year are when extra weight matters most.

- Age, at both ends. A puppy can earn a lifetime of lowered heat tolerance through one negligent day. For an older dog it can simply be deadly.
- Muzzle length. The shorter-faced breeds, the brachycephalics, can't dump heat as easily. Pay special attention here.
- Laryngeal paralysis. Common in older retrievers and other large breeds, and a substantial risk factor in its own right. If your dog has been diagnosed with it, or has a hoarse change in his bark, or the noisy breathing called stridor that can come before a diagnosis, treat every warm day as a dangerous one. These dogs do not tolerate heat stress, because their airway is compromised, and they are at far greater risk of heat injury and suffocation. They need to be retired from strenuous work.
- A previous heat episode. Think of a dog's thermoregulation as a thermostat that can't be replaced. If it blows once, it can lower his tolerance for heat for the rest of his life.
- Drive. The dog that will not quit is the dog most likely to be in trouble before you notice. If you're reading this guide, your dog might be in that crowd. Managing it is your job.

FIELD NOTE

If you remember three things from this chapter, make them these: poor conditioning, excess drive, and any previous heat injury. A dog with laryngeal paralysis should already have been retired, and an overweight dog, a dark coat and an old dog are all things you can see. Conditioning, drive and history are the three that hide.



RECOGNITION

The warning signs, in the order they appear

Heat injury is a sequence of events, not a single moment. If you know where your dog is on this list, you know what you should be reacting to.

The reason for putting these in order is that the early signs are easy to talk yourself out of. Knowing what follows them is what gives them the weight they deserve.

One note on temperature before you start. A thermometer reading on its own will not tell you which stage your dog is in. The signs will. That is why this list is built around what you can see.

Early: heat stress

WHAT YOU SEE

- Heavy panting that is still under his control. He can stop when he wants to, to sniff or to swallow.
- A tongue wider and longer than usual.
- A faster heart rate, with a pulse you can still feel clearly inside the thigh.
- Temperature climbing, past 103 degrees.
- He goes looking for shade, lies flat out on any cool surface he can find, hunts for water and drinks hard.
- Small lapses in attention. An example Charles sees in his own dogs is a missed come-around whistle, and a dog casting wider and wider as if running in a daze, not answering commands he knows cold.

WHAT IT MEANS

This stage is easy to dismiss as disobedience or poor conditioning. It is neither. Heat stress corrects itself when the heavy exercise stops, which is exactly why it is so easy to push through.

WHAT YOU DO

Stop the work. Shade, moving air, water. Do not finish the field.

Serious: heat exhaustion

Think of this as moderate heat injury, and know that it is a short trip from here to a true emergency and a potentially poor prognosis.

WHAT YOU SEE

- Panting he can no longer control, sometimes called air hunger, with loud breathing. He can't stop panting long enough to drink or take food, because every effort is going into shedding heat.
- A heart rate well up, above 140 to 160 even after he has stopped working, and a pulse inside the thigh that has gone weak.
- Gums turning dark, brick red rather than pink. Press a finger against the gum and let go. The color should come back in under two seconds. Two to three seconds or longer is a warning.
- Disorientation. He can no longer take commands or do his job.
- Heavy drooling that has turned into a dry mouth.
- Nausea, vomiting or diarrhea.
- Weakness, and possibly collapse.

WHAT YOU DO

A dog at this stage may come back on his own once the work stops. Don't count on it. It is far better to recognize the danger and start cooling now. Everything in the next chapter applies.

Critical: heatstroke

Now you are in trouble.

Heatstroke is often defined as a rectal temperature above 106 degrees. But studies of police and military dogs and search and rescue dogs have recorded temperatures well above 106 while they were working, in dogs that were completely unaffected and recovered quickly once the work stopped. That shows how much fitness and acclimation matter. It also means a temperature reading alone does not tell you a dog has heatstroke. The signs do.

WHAT YOU SEE: EVERYTHING ABOVE, PLUS

- Mental incapacitation: dullness, stupor, uncontrolled shaking or tremors, seizures, coma.

- Breathing effort and rate severely increased, with very noisy breathing.
- A poor pulse inside the thigh.
- Vomiting and diarrhea, either of which may contain blood.

WARNING

A falling temperature is not always good news. As circulation fails, rectal temperature can drop. A dog with these signs whose temperature is coming down may be getting worse, not better. Keep going to the vet and continue to monitor after he's been treated.

WHAT COMES NEXT

- Disseminated intravascular coagulation, or DIC. The blood loses its ability to clot, and the dog bleeds spontaneously: from the nose, mouth, gut, urinary tract, lungs, skin, eyes, even the joints.
- Multiple organ failure, especially the kidneys, liver, gut and heart. Some of it may not show for 12 to 72 hours after the episode. This is why a dog that seems to recover in the field should still be seen by a veterinarian.
- Cardiac complications, including severe arrhythmias and cardiopulmonary arrest. Fluid in the lungs is also possible.
- Severe brain injury.
- Death.

The first five minutes

If your dog is suffering acute heat injury, whether heat stress or heatstroke, be prepared to act. What you do before the truck moves matters more than the drive. This is also where some of the advice in circulation is out of date, and it is one of the reasons this guide exists.

Lower his temperature in the field as quickly as you possibly can. As one critical care veterinarian put it, once the body's cells are cooked, they cannot be made raw again.

Do this

- 1 Get him out of the sun. Find shade, or make it.
- 2 Take off his vest, and any tight collar or boots.
- 3 Get air moving across him. A moving vehicle with the windows down, or a fan on high. Charles has used helicopter rotor wash on overheated elk in the mountains during research. Use whatever you have.
- 4 Use cold water, or ice. Put it where the hair is thinnest: the armpits, the groin, the inside of the ear flaps and the belly.
- 5 If he can keep his head up and protect himself from drowning, put him in cool water. A stream, a lake.
- 6 Cold water or isopropyl alcohol on the feet.
- 7 Offer cool water to drink if he is able to take it.

A word on cold water, because this is where the old advice is wrong. For years the argument was that cold water on the skin constricts the blood vessels near the surface and traps heat in the core. That effect has more recently been argued to be insignificant, and cold water or ice brings core temperature down more effectively. This is the current recommendation, and it contradicts a lot of what is still being written. Ice water baths are the first line of defense in human heatstroke.

A thick-coated dog can be further insulated if you soak the coat down. If you do, use cold water and get a fan on him. In a hospital, long and thick-coated dogs are sometimes shaved.

Do not do this

- Do not wait to do something.

- Do not put an unconscious dog in a lake or pool without keeping his head out of the water.
- Do not force him to drink. He is likely to inhale the water and develop pneumonia.
- Do not overshoot into hypothermia. Stop cooling when his rectal temperature reaches 103 degrees.
- Do not drape wet towels over him and leave them there. They become insulating blankets and hold the heat in. Towels can work on a short-coated dog, but only with cold water, and only if you keep recharging them.

How long to cool

That depends on where you are. Two miles from the truck, cooling a dog in a creek may be your best option. Ten to twenty minutes is a reasonable amount of time to see measurable cooling effects.

Stop cooling when his temperature reaches 103 degrees. Without a thermometer you won't know when that is, which is the best argument there is for carrying one. If he starts shivering, stop.

If you can reach a vehicle quickly, cool him in the field for about ten minutes and then move him. Your air-conditioned vehicle may be one of the best tools you have. If you're not alone, have someone bring the vehicle to you while you work on the dog, and get it cooled down before he goes in.

Stop and drive now if

- He has a seizure.
- His heart rate is dropping quickly, or his pulse is getting weak.
- He shows any sign of heatstroke: stupor, tremors, collapse, or blood.
- His temperature has come down to 103. Stop cooling and drive.
- Ten to twenty minutes of cooling hasn't turned him around.
- You are not sure.

The drive, and the call

Here is the reality. If your dog has severe heatstroke, you do what you can in the field to bring his temperature down as fast as possible, but you are not the definitive care or the last care he will need. Get him to a veterinarian as quickly as you can, and understand that the outcome can stay undetermined, sometimes for several days.

Know before you go how far you are from emergency veterinary care, and how long it takes to get there. Heatstroke, like bloat, warrants stopping at the first veterinary hospital you can find. It may not be your first choice or your last stop, but triage care can change the outcome.

When you decide to drive, call the hospital on the way so they can be ready for you and give you any further guidance.

ON THE PHONE

I'm on my way with a working dog I think has heatstroke. He's a [breed], [age], with a [coat type] coat, and he's [in condition / not in condition]. Right now he's [what you're seeing]. I've been cooling him with [what you did] for [how long], and his temperature was [reading / I don't have one]. I'm [minutes] away. What do you want me to do on the way?



THE PART NOBODY WRITES ABOUT

Training days, test days, truck heat, and waiting

Some heat emergencies in working dogs don't happen while the dog is running. They happen in a vehicle, in a crate. The morning's shade moved, or forty minutes passed between the dog finishing a field and anyone looking at him again.

For those of us who run in hunting test formats, late August and September is test season across the Northeast and the Midwest, which means a lot of our dogs are crated in trucks in the hottest part of the year. Handlers are nervous about running order, not thinking about a crate, and there is a lot of waiting.

The temperature inside a closed vehicle can climb to dangerous levels in a very short time. Take every precaution.

Before you leave the house

- Bring water for the dog, separate from the water you plan to drink.
- Consider baiting his water with hydration tablets or FortiFlora. It often gets a dog to drink more.
- Plan the parking at your destination, not just the route. Where will the shade be at noon, not at eight?
- Bring a pop-up tent. Fully opened over the back of the truck, it can throw a lot of shade even in a sunny spot.
- Pack a few squeeze clamps and a shade cloth or tarp. Draped over the truck cap and clamped to the open windows and rear glass, it gives you shade and keeps the air moving.

- Invest in airflow for the kennel. Manufacturer-specific fans are a clean option, but battery fans from the tool companies often run longer and move more easily.
-

Between runs

- Airflow beats shade, and both beat neither.
 - A crated dog in a parked vehicle is a different situation from a dog on a chain in the shade. Know which one you have, and which is better for your dog.
 - If you're staking your dog out, slide a bundt pan down the stake before you drive it in, then fill it with water. An unattended dog can't knock it over or empty it.
 - Keep water available at all times. If you can't, water him from a squeeze bottle often.
 - Consider two cooling vests, one on ice in a cooler while the other is in use, then swap them. Don't leave one on too long, or it becomes an insulating blanket.
-

The drive home

- A dog that has worked hard in the heat is not out of the woods when the tailgate closes.
- Before you crate him, make sure he can fully close his mouth now and then. It's a general sign he is cooling down safely.
- Dry him off as much as you can.
- Above all, if you're concerned, put the dogs in the cab with you and turn up the air conditioning. A dirty truck cab is a small price to pay to make sure your dog makes it to the next covey of birds. He would do it for you.

Water, food, and the road

Electrolytes

The evidence shows electrolyte products are not useful for canine athletes. The reason is simple: dogs don't sweat the way people do, so they lose very little in the way of electrolytes when they work. What matters is water.

Feeding

On particularly hot days, or after an unusual heat stress episode, it may be best to hold off feeding longer than usual. Heat stress can injure the gut, and adding the heat of digesting a meal to a dog that is already struggling doesn't help. Food comes later. Hydration comes first.

AFTERWARD

The next 48 hours, and the rest of the season

Even a less severe heat injury can cause problems later. Having your dog seen by a veterinarian to check his organ function is wise, even if he seems fine. Remember that some organ damage may not show for 12 to 72 hours.

OVER THE NEXT FEW DAYS, WATCH FOR

- Vomiting or diarrhea, especially with blood.
- Drinking and urinating noticeably more than usual.
- Urinating dramatically less than usual.

And remember the thermostat. A single episode can lower a dog's tolerance for heat for the rest of his life. Treat him as a dog with a smaller margin from here on.

What this protocol actually requires

A protocol is just paper if you don't have it with you, or don't know how to use it. Have it in the truck, or none of this matters. Everything this guide asks of you comes down to a short list:

- A rectal thermometer, and lubricant. Without one, stopping at 103 is guesswork.
- More water than you think you need, some of it cold.
- A cooler with ice.
- A way to move air: a battery fan.
- A way to make shade: a pop-up tent, or a tarp and clamps.
- Towels, used and recharged, never left on.
- Your vet's number and the nearest emergency clinic, written down before you leave.

What this guide is not

This is general first-aid information for people who work with canine athletes. It is not a diagnosis, it is not a treatment plan, and reading it does not create a veterinarian-client-patient relationship with the authors.

You and your veterinarian know your dog. This guide does not. When the two disagree, listen to your veterinarian.

Write the name and number of your own vet, and of the nearest emergency clinic to wherever you're hunting, on the protocol card in your Brushworn field health kit before you need them. Looking up a phone number is not something you want to be doing when it counts. And let's face it, cell service can be unreliable, if not nonexistent, in our favorite places.